

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-2731D0C8 2731d0c8-1059-4c35-be07-c86b45a90c7a
Submitted by	testreviewer L!
Submission date	2026-10-07 06:44 UTC
Documents	1 Types: Not yet classified
Case type	School document
Status / decision	Submitted / Pending review
Report generated	2026-10-07 06:52 UTC by testreviewer L! (Reviewer L1)
Report ID	52dd4791-adff-48f8-9927-6b8fba27188f

Preliminary report. Automated analysis had not finished when this report was generated. Still outstanding: Text extraction and classification of 'case11School educational document.pdf'.

2. Executive Summary

Not yet scored. The risk engine has not produced an assessment for this case, so there is no overall tier.

Why: No risk tier yet — automated analysis is still in progress.

Decision status: Pending review.

Exceptions: 2 item(s) did not pass, 2 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The case has not been scored, so there are no triggered risk reasons yet.

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — case11School educational document.pdf (Not yet classified · 3 passed, 2 flagged)

Text extraction had not finished when this report was generated.

Check	Result	Details
Error level analysis (ELA) Pixel-level recompression differences that can indicate edited regions.	Flag	1 finding(s). Page 1: possible anti-forensic smoothing detected (noise variance=4.47) — could indicate an attempt to hide edit traces.
Copy-move detection Regions of a page that were duplicated elsewhere on the same page.	Pass	No anomaly detected by this check.
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Flag	1 finding(s). Page 1 — Resolution / sharpness consistency: The signature and the circular stamp at the bottom right are noticeably softer and less sharp compared to the printed text, which is consistent with natural pen and stamp ink. This is a vision-model j...
Signature / stamp detection Whether a signature or stamp mark is present where this kind of document normally carries one.	Pass	2 signature/stamp region(s) located (presence only).
Duplicate detection Perceptual-hash comparison of each page against previously submitted pages.	Pass	No anomaly detected by this check.

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (2 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where	Highlight
1	ELA: possible anti-forensic signal Page 1: possible anti-forensic smoothing detected (noise variance=4.47) — could indicate an attempt to hide edit traces.	Document 1 (case11School educational document.pdf) · page 1	R1
2	Visual review: resolution / sharpness consistency inconsistency (model judgment) Page 1 — Resolution / sharpness consistency: The signature and the circular stamp at the bottom right are noticeably softer and less sharp compared to the printed text, which is consistent with natural pen and stamp ink. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definiti...	Document 1 (case11School educational document.pdf) · page 1	R2

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Did not run	Did not run on any of the 1 document(s).
Font consistency	Did not run	Did not run on any of the 1 document(s).
Error level analysis (ELA)	Ran	Ran on 1 of 1 document(s); 1 flagged.
Copy-move detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 1 flagged.
Signature / stamp detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Duplicate detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Field validation	Did not run	Did not run on any of the 1 document(s).
Issuer verification	Did not run	Did not run on any of the 1 document(s).
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

This case had not been scored when the report was generated, so no risk-rule versions apply.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.



7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-07 06:44 UTC	Case created	testreviewer L! (Reviewer L1)	
2026-10-07 06:44 UTC	Document uploaded	testreviewer L! (Reviewer L1)	case11School educational document.pdf; SHA-256 4753e0376fdd7679...
2026-10-07 06:48 UTC	Visual inconsistency review completed	System	case11School educational document.pdf
2026-10-07 06:48 UTC	Signature / stamp detection completed	System	case11School educational document.pdf
2026-10-07 06:49 UTC	Duplicate detection completed	System	case11School educational document.pdf
2026-10-07 06:49 UTC	ELA and copy-move checks completed	System	case11School educational document.pdf

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — case11School educational document.pdf

No fields were extracted from this document.

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256	Verification
case11School educational document.pdf 122,398 bytes	4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a	VERIFIED

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
Colour model	RGB, converted to BGR for OpenCV

Error level analysis (ELA)

JPEG recompression quality	75
Error gain (scale)	50 (sqrt error term; the reference tool's default)
Contrast stretch / LUT	not applied — the raw error magnitude is thresholded directly
Region threshold	page-relative percentile 99.7 with an absolute floor of 60/255
Region area accepted	0.2% – 60% of the page
Applies to	pages with an embedded raster image only (born-digital text pages are exempt)

Copy-move detection

Keypoint algorithm	BRISK (OpenCV), detector threshold 200
Response / matching / distance thresholds	90 / 20 / 15
Minimum cluster size	30 matching keypoint pairs
Minimum region area	2% of the page
Keypoint / match caps	4,000 / 4,000

Duplicate detection

Hash	perceptual hash (imagehash pHash), 64-bit, one per rendered page
Hamming-distance threshold	5 (≤ this is flagged)
Compared against	every page hash of every previously submitted document

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge ≤ 1536px

Signature / stamp detection and comparison

Model	gpt-4.1-mini
Pages analysed per document	first 10
Output	presence and placement only; comparison is a qualitative verdict, never a numeric score



9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

	ELA (solid red)
	Copy-move (solid orange)
	Font mismatch in the text layer (solid fuchsia)
	Field exception (solid purple)
	Model judgment (dashed blue, approximate)
	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (case11School educational document.pdf)	1	ELA — possible anti-forensic signal
R2	Document 1 (case11School educational document.pdf)	1	Visual review — resolution / sharpness consistency (approximate)

case11School educational document.pdf — page 1

Section 9 · Document 1 of 1 · regions R1, R2

 Model judgment (dashed blue, approximate)  Page-level finding (not localized)

مؤسسة الإمارات للطاقة النووية
Emirates Nuclear Energy Corporation

P.O. Box 112010, Abu Dhabi, United Arab Emirates
Telephone: +971 2 3150 555 Fax: +971 2 3150 556 www.enec.gov.ae

Reference Number : SC20231027/23051672/00001631

20 Dec 2023

Emirates NBD
Abu Dhabi - United Arab Emirates

Salary Certificate

This is to certify that under mentioned person is currently employed at Emirates Nuclear Energy Corporation located in Emirate of Abu Dhabi.

Name	: Nagasurya Nageshwaran Nageswaran
Nationality	: Indian
Date of Join	: 10 Jan 2023
Employee ID	: 00001631
Work Location	: BNPP - Barakah
Total Fixed Salary	: 44094.75 AED
Total Variable Salary	: 6300.00 AED
Total Lump Sum	: 50394.75 AED/Month
Position	: IT Engineer
Grade	: 18
Shift Pattern	: MON2FRI

Note that his monthly salary will be transferred to Abu Dhabi Commercial Bank to the following account number 1312422920001, AE900030013124242920001.

This certificate has been issued upon the person's request without any liability or obligation whatsoever on the part of Emirates Nuclear Energy Corporation. The certificate shall remain valid for thirty calendar days from the date of issue. Any alteration or modification to this certificate shall render the certificate null and void.

Yours Sincerely

Ahmed Jasem AlShamsi
Enterprise Chief Human Capital Officer

*Variable allowances are subject to change

Note: This is a system-generated certificate bearing a generated fully authentic non-physical signature. It shall be considered as original as its issue does not require any further ENEC approval.



R1. Error level analysis (ELA) (page-level — not localized to a region)

Page 1: possible anti-forensic smoothing detected (noise variance=4.47) — could indicate an attempt to hide edit traces.

R2. Visual inconsistency / AI-generation review

Page 1 — Resolution / sharpness consistency: The signature and the circular stamp at the bottom right are noticeably softer and less sharp compared to the printed text, which is consistent with natural pen and stamp ink. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Not counted: it is about the detected signature (covers 100% of it) — pen strokes and ru...